

No. of Functional Features Included in the Solution

Date	5 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

The solution includes the following functional features which enable users to upload or record audio explanations, analyze conceptual understanding using NLP, and provide feedback, scoring, and reports through a web application.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Remarks / Justification
1	User Authentication	Allows users to register, login, logout and reset password.	Auth Service	Done	Secure access for students, teachers and admins.
2	Audio Upload / Recording	Users can upload an audio file or record directly from the application.	Audio Service	Done	Supports mp3, wav, m4a formats.
3	Speech-to-Text Transcription	Converts uploaded/recorded audio to text using STT API (Whisper).	STT Service	Done	High accuracy transcription.
4	NLP Analysis & Concept Matching	Analyzes the transcript, extracts concepts and matches with ideal reference answers.	NLP Service	Done	Uses embeddings and semantic similarity.
5	Scoring & Evaluation	Generates score based on concept coverage, accuracy and completeness.	Scoring Engine	Done	Weighted scoring algorithm.
6	Feedback & Suggestions	Provides detailed feedback with strengths, gaps and improvement suggestions.	Feedback Engine	Done	Helps in learning improvement.
7	Dashboard & Reports	Displays summary, scores, feedback, history and progress charts.	Reporting Module	In Progress	Visual analytics in development.
8	Admin Panel	Manage users, questions, reference answers and view system logs.	Admin Module	In Progress	Role-based management.
9	Export Report	Allows users to export feedback and reports as PDF.	Report Service	Pending	To be added in next sprint.
10	Notifications	Sends email notifications for reports and updates.	Notification Service	Pending	Email via SMTP integration.

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented (Done)	6
Features In Progress	2
Pending / Not Yet Implemented	2
Features Tested & Verified	6

Additional / Nice-to-have Features	Multi-language support, Advanced analytics dashboard
Features Deferred (If Any)	Real-time collaboration, Mobile app

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Management	2	User Authentication, Admin Panel
2	Input & Processing	3	Audio Upload/Recording, Speech-to-Text, NLP Analysis
3	Evaluation & Output	3	Scoring & Evaluation, Feedback & Suggestions, Export Report
4	Analytics & Reporting	1	Dashboard & Reports
5	Security & Communication	1	Notifications